

ECE 607

Homework #2

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Part 1:

1)

$$P(x_i | \mu) = \mu^{x_i} (1 - \mu)^{1 - x_i}$$

$$L(\mu) = \prod_{i=1}^N P(x_i | \mu)$$

$$= \prod_{i=1}^N \mu^{x_i} (1 - \mu)^{1 - x_i}$$

$$= \ln L(\mu) = \sum_{i=1}^N \ln(\mu^{x_i} (1 - \mu)^{1 - x_i})$$

$$L(\mu) = \left(\sum_{i=1}^N x_i \right) \ln \mu + \left(\sum_{i=1}^N (1 - x_i) \right) \ln (1 - \mu)$$

$$\frac{d}{d\mu} L(\mu) = \frac{1}{\mu} \sum_{i=1}^N x_i - \frac{1}{1 - \mu} \sum_{i=1}^N (1 - x_i) = 0$$

$$\frac{1}{\mu} \sum_{i=1}^N x_i = \frac{1}{1-\mu} \sum_{i=1}^N (1-x_i)$$

$$N_1 = \sum_{i=1}^N x_i \quad N_0 = \sum_{i=1}^N (1-x_i)$$

$$\frac{N_1}{\mu} = \frac{N - N_1}{1-\mu}$$

$$N_1(1-\mu) = \mu(N - N_1)$$

$$N_1 - N_1\mu = N\mu - N_1\mu$$

$$N_1 = N\mu + \cancel{N_1\mu} - \cancel{N_1\mu}$$

$$N_1 = N\mu$$

2)

likelihood

$$P(D|\mu) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)}$$

Prior

$$P(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)}$$

$$P(\mu|D) \propto P(D|\mu)P(\mu)$$

$$\ln P(\mu|D) = \sum_{i=1}^N \ln P(x_i|\mu) + \ln P(\mu) + \text{const.}$$

$$\ln P(\mu|D) = -\frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2 - \frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 + \text{const.}$$

$$\frac{\partial}{\partial \mu} \ln P(\mu|D) = \frac{1}{\sigma^2} \sum_{i=1}^N (x_i - \mu) - \frac{\mu - \mu_0}{\sigma_0^2} = 0$$

$$\frac{1}{\sigma^2} \left(\sum_{i=1}^N x_i - N\mu \right) - \frac{\mu - \mu_0}{\sigma_0^2} = 0$$

$$\sigma_0^2 \left(\sum_{i=1}^N x_i - N\mu \right) - \sigma^2 (\mu - \mu_0) = 0$$

$$\mu_{MAP} = \frac{\sigma_0^2 \sum_{i=1}^N x_i + \sigma^2 \mu_0}{N\sigma_0^2 + \sigma^2}$$

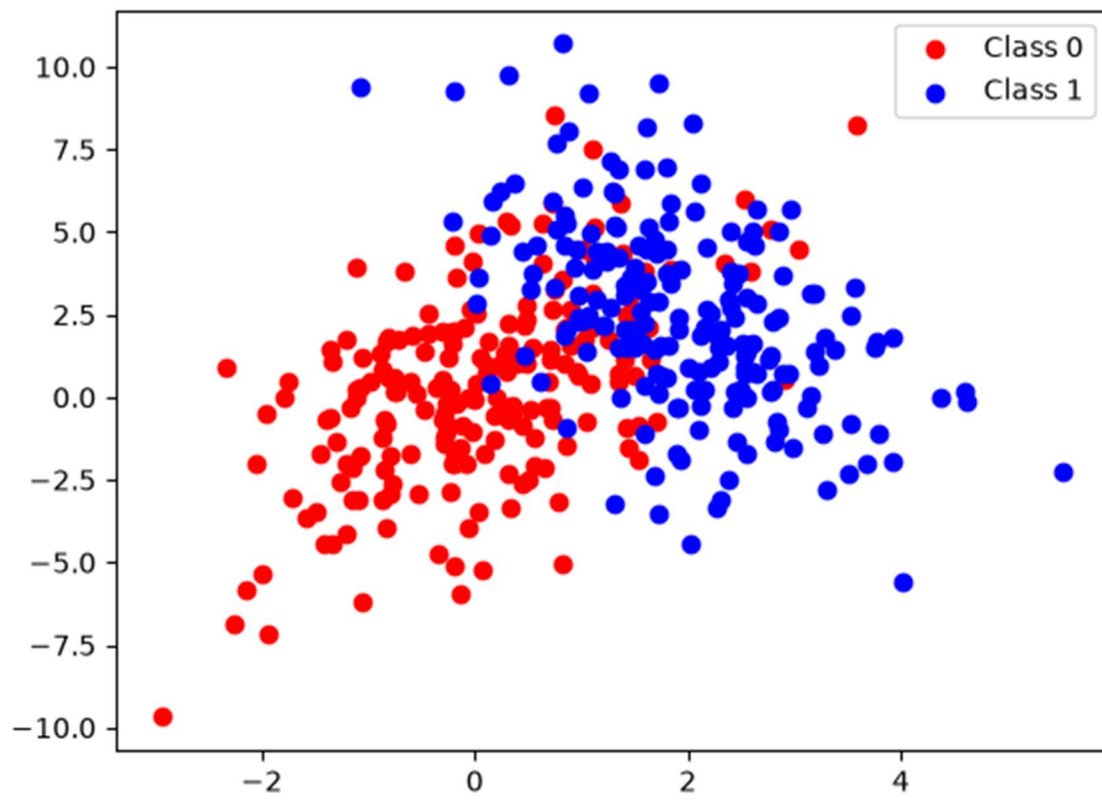
$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\tau = \frac{1}{\sigma^2} \quad \tau_0 = \frac{1}{\sigma_0^2}$$

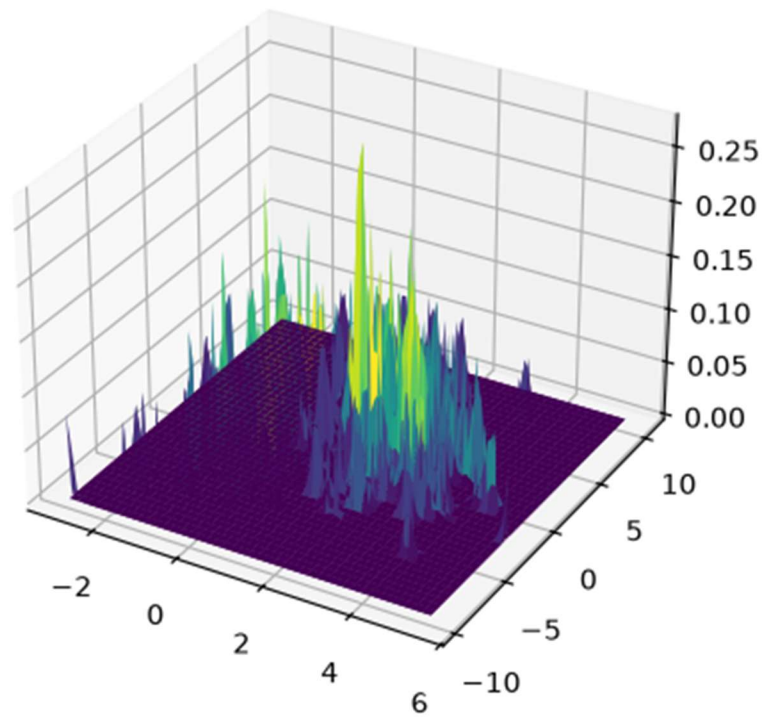
$$\mu_{MAP} = \frac{N\tau\bar{x} + \tau_0\mu_0}{N\tau + \tau_0}$$

Part 2:

Task 1)

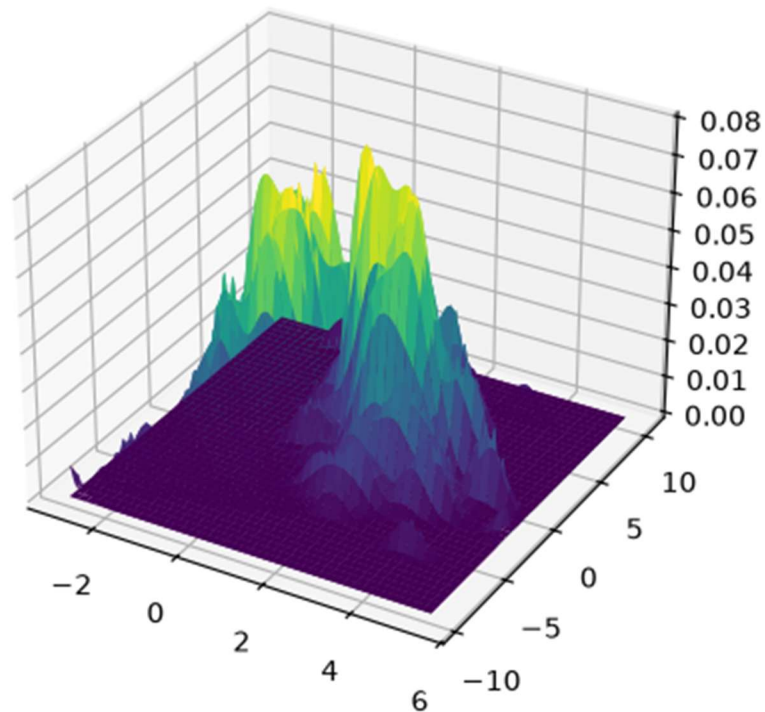


Non-parametric method $h=0.1$



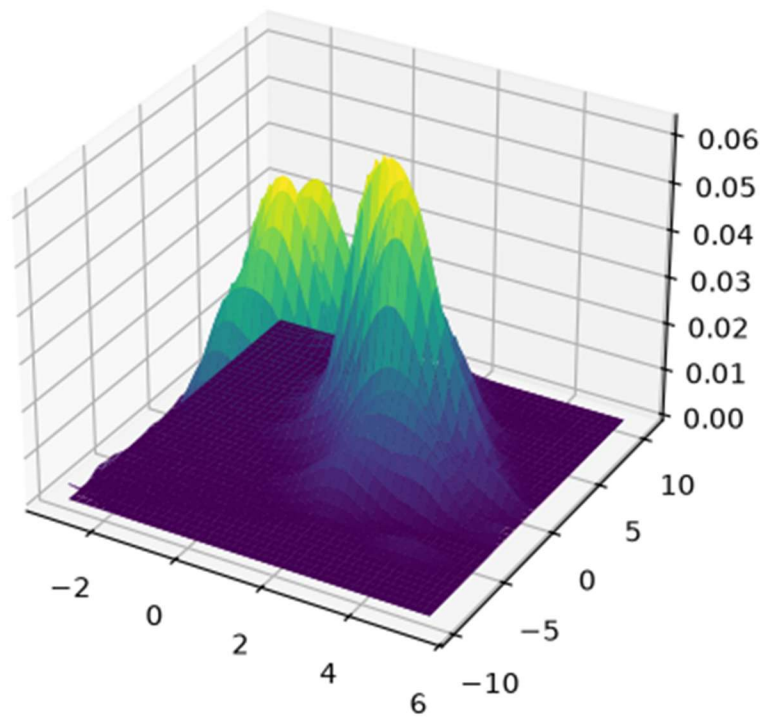
For $h = 0.1$, the density estimate is very sharp and contains many narrow peaks around individual data points.

Non-parametric method $h=0.3$



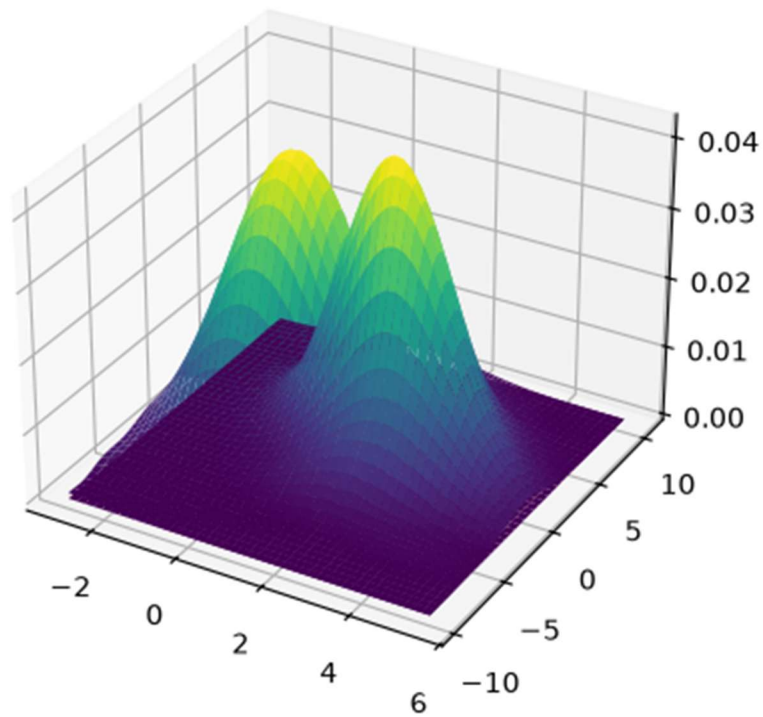
For $h = 0.3$, the density becomes smoother while still preserving most of the important structure of the data. The individual peaks are less pronounced than with $h = 0.1$.

Non-parametric method $h=0.5$



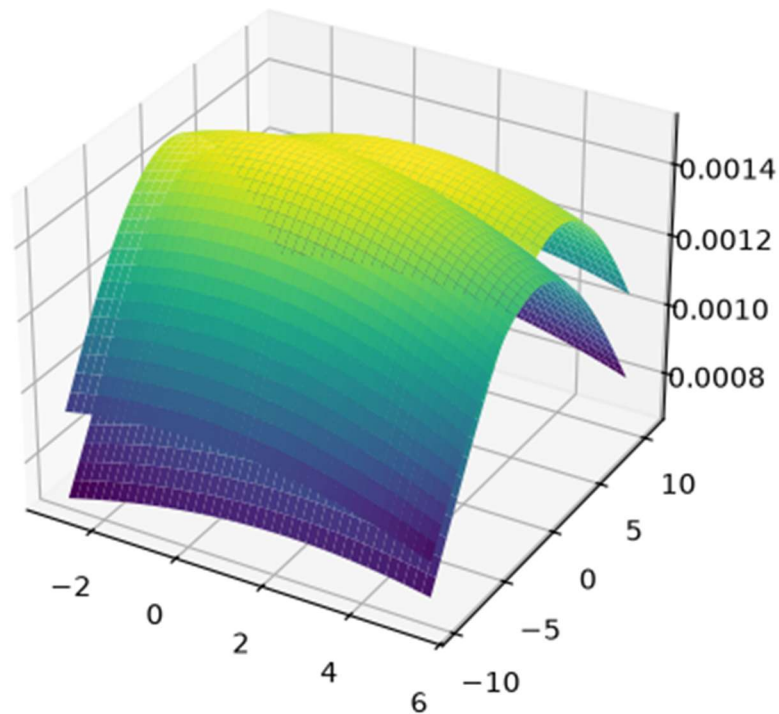
For $h = 0.5$, the density is smooth while maintaining the main peaks. I see this being the best fit for the data.

Non-parametric method $h=1$



For $h = 1$, the density is smoother than for $h = 0.5$. The structure of the data is still visible, although some details are lost.

Non-parametric method $h=10$



For $h = 10$, the density is very smooth, and the individual peaks are lost.

As h increases, the smoothing in the estimated density increases. Meaning a smaller h is less smooth but more detail and a larger h is smoother but less detail.

MODELS	Accuracy
Parametric	87.5%
Naive Bayesian	88%
Non-parametric $h = 0.1$	83.5%
Non-parametric $h = 0.3$	87.75%
Non-parametric $h = 0.5$	88.5%
Non-parametric $h = 1$	88%
Non-parametric $h = 10$	72.75%
KNN $k = 1$	82.25%
KNN $k = 5$	87.75%
KNN $k = 15$	87.5%
KNN $k = 155$	82%

The results show that the non-parametric classifier with $h = 0.5$ performed best, achieving 88.5% accuracy. The parametric classifier also performed well with 88%, while the Naive Bayesian classifier achieved 87.5%. For the non-parametric method, the kernel size has a noticeable effect. A small value such as $h = 0.1$ produced a less accurate result of 83.5%, while $h = 0.5$ gave the best result. The large value $h = 10$ resulted in only 72.75% accuracy as the density estimate becomes too smooth and loses important features of the data. For K-NN, $k = 5$ performed best with 87.75% accuracy. Very small or very large values of k performed worse as they are too sensitive to individual points. Overall, $h = 0.5$ provided the best fit for the non-parametric density estimate and achieved the highest accuracy of 88.5%.